MA 324
STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS
IIT GUWAHATI

END-SEMESTER EXAMINATION
14:00–17:00 IST
MAY 8, 2026

PART-A

Name of the Student: N. Rikhith Reddy

Signature of the Student: Rikhith Roll No.: 230123038

Signature of the Invigilator: Gaurav Kr. Meher

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3	9	17

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

writing in matrix forms

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

$$Y = X\theta + \varepsilon$$

Y - response variable
X - independent variable
θ - Parameters
ε - Error

as we know the LSE estimator of θ is $\hat{\theta} = (X^T X)^{-1} X^T Y$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$\det(X^T X) = 6 \times 6 - (3)^2 = 36 - 9 = 27$$

$$(X^T X)^{-1} = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(X^T X)^{-1} X^T = \cancel{\frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}} = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$\hat{\theta} = (X^T X)^{-1} X^T Y$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$$

$$\hat{\theta} = \frac{1}{3}\begin{bmatrix} y_1 + y_3 \\ y_1 - y_2 \end{bmatrix}$$

$$\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$$

$\Rightarrow$ $\hat{\theta}_1 = \dfrac{y_1 + y_3}{3}$, $\hat{\theta}_2 = \dfrac{y_1 - y_2}{3}$ are the LSE estimators of θ_1, θ_2 respectively.

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2. Let U_1 and U_2 be two independent uniform random variables on the interval (0, 1). Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

a) Given $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, U_1, U_2 – independent uniform random variables

correlation matrix

$$\rho(X) = \begin{bmatrix} \rho(X_1,X_1) & \rho(X_1,X_2) & \rho(X_1,X_3) \\ \rho(X_2,X_1) & \rho(X_2,X_2) & \rho(X_2,X_3) \\ \rho(X_3,X_1) & \rho(X_3,X_2) & \rho(X_3,X_3) \end{bmatrix}$$

$\rho(X_1,X_1) = \rho(X_2,X_2) = \rho(X_3,X_3) = 1$

$\rho(X_1,X_2) = \rho(U_1,U_2) = 0$ since the both U_1 and U_2 are independent

given $X_1 = U_1 \Rightarrow$ ~~$E(X_1) = 0$, $Var(X_1) = 1$~~ as $U_1\ U_2$ are uniform random variables
$X_2 = U_2 \Rightarrow$ ~~$E(X_2) = 0$, $Var(X_2) = 1$~~

$X_3 = U_1 + U_2$

which is a linear combination of U_1 & U_2

$E(X_3) = E(U_1+U_2) = E(U_1) + E(U_2) = 0 + 0 = 0$

$Var(X_3) = Var(U_1+U_2) = Var(U_1) + Var(U_2) + 2\,cov(U_1, U_2)$

$= 1 + 1 + 0$ $\quad U_1\ U_2$ are independent

$= 2$

$$\rho(X_1, X_3) = \frac{cov(X_1, X_3)}{\sqrt{Var(X_1)}\cdot\sqrt{Var(X_3)}} = \frac{cov(U_1, U_2+U_1)}{\sqrt{1}\cdot\sqrt{2}} = \frac{cov(U_1, U_1+U_2)}{\sqrt{2}}$$

applying $cov(X,Y) = E\left([X-E(X)][Y-E(Y)]\right)$

$$= \frac{E\left([U_1 - E(U_1)][U_1+U_2 - E(U_1+U_2)]\right)}{\sqrt{2}}$$

$E(U_1) = 0$

$E(U_1+U_2) = E(U_1) + E(U_2) = 0$

$$= \frac{E(U_1(U_1+U_2))}{\sqrt{2}}$$

$$= \frac{E(U_1^2 + U_1U_2)}{\sqrt{2}}$$

$$= \frac{E(U_1^2) + E(U_1U_2)}{\sqrt{2}}$$ $\quad U_1 U_2$ are independent

$E(U_1^2) = E((U_1 - 0)^2)$

$= E((U_1 - E(U_1))^2)$

$= Var(U_1)$

$$= \frac{Var(U_1) + E(U_1)E(U_2)}{\sqrt{2}}$$

$\rho(X_1, X_3) = \frac{1}{\sqrt{2}}$

Similarly

$\rho(X_2, X_3) = \frac{\operatorname{cov}(X_2, X_3)}{\sqrt{\operatorname{var} X_2} \cdot \sqrt{\operatorname{var} X_3}} = \frac{\operatorname{cov}(U_2, U_1+U_2)}{\sqrt{1}\cdot\sqrt{2}} = \frac{E\left([U_2 - E(U_2)][U_1+U_2 - E(U_1+U_2)]\right)}{\sqrt{2}}$

$= \frac{E(U_2(U_1+U_2))}{\sqrt{2}}$

$= \frac{E(U_2^2 + U_1U_2)}{\sqrt{2}}$

$= \frac{1}{\sqrt{2}}$

Correlation matrix ρ of X is

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$$\rho(X) = \begin{bmatrix} \rho(X_1,X_1) & \rho(X_1X_2) & \rho(X_1X_3) \\ \rho(X_2X_1) & \rho(X_2,X_2) & \rho(X_2X_3) \\ \rho(X_3X_1) & \rho(X_3X_2) & \rho(X_3X_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

b) Variance of first principal component is equal to the largest eigen value of the correlation matrix from part (a)

eigen values of $\rho(X)$ are λ s.t

$$|\rho(X) - \lambda I_3| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} = 0$$

$(1-\lambda)\left[(1-\lambda)^2 - \frac{1}{2}\right] + \frac{1}{\sqrt{2}}\left(-(1-\lambda)(\tfrac{1}{\sqrt{2}})\right) = 0$

$(1-\lambda)\left[1+\lambda^2-2\lambda-\frac{1}{2}\right] - \frac{1}{2}(1-\lambda) = 0$

$(1-\lambda)\left[\cancel{1}+\lambda^2-2\lambda-\cancel{\tfrac{1}{2}}-\cancel{\tfrac{1}{2}}\right] = 0$

$(1-\lambda)(\lambda^2-2\lambda) = 0$

$(1-\lambda)\,\lambda(\lambda-2) = 0$

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$\Rightarrow \lambda = 0, 1, 2$ are the eigen values, largest is 2

$\Rightarrow$ Variance of first principal component is 2.

3. Let the matrix of distance be

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & & & \\ 2 & 1 & 0 & & \\ 3 & 11 & 2 & 0 & \\ 4 & 5 & 3 & 3.5 & 0 \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:
   i. Complete linkage hierarchical procedure.
   ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) complete linkage heirarchical procedure

$d_{(uv)w} = \max\{d_{uw}, d_{vw}\}$

Given distance matrix

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

distance $\propto \frac{1}{\text{similarity}}$. at each step we group the items with high similarity i.e min distance

min is $d_{12} = 1$ – $\{1,2\}$ will be grouped

$d_{3,12} = \max(d_{31}, d_{32}) = \max(11, 2) = 11$

$d_{4,12} = \max(d_{41}, d_{42}) = \max(5, 3) = 5$

$d_{3,4} = 3.5$

↓

	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0

min is $d_{34} = 3.5$ – $\{3,4\}$ will be grouped

~~$d_{12,34} = \max(d_{1,34}, d_{2,34}) = \max($~~

$d_{34,12} = \max(d_{3,12}, d_{4,12}) = \max(11, 5) = 11$

↓

	(12)	(34)
(12)	0	
(34)	11	0

~~the~~ (12) (34) only two are left so finally they will grouped $\{1,2,3,4\}$

↓

	(1234)
(1234)	0

Grouping order is

$\{1,2,3,4\} \rightarrow \{\{1,2\},3,4\} \rightarrow \{\{1,2\},\{3,4\}\} \rightarrow \{\{1,2,3,4\}\}$

iii) average linkage hirarchial procedure

$$d_{(uv)w} = \frac{\sum d_{uw} + \sum d_{vw}}{N_{uv} * N_w}$$

Given distance matrix

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

min is $d_{1,2}=1$ — {1,2} will be grouped

$d_{3,4} = 3.5$

$d_{12,3} = \frac{d_{13}+d_{23}}{2} = \frac{11+2}{2} = 6.5$

$d_{12,4} = \frac{d_{14}+d_{24}}{2} = \frac{5+3}{2} = 4$

↓

	(12)	3	4
(12)	0		
3	6.5	0	
4	4	3.5	0

min is $d_{34} = 3.5$ — {3,4} will be grouped

$$d_{12,34} = \frac{d_{1,3}+d_{1,4}+d_{2,3}+d_{2,4}}{2 * 2}$$

$$= \frac{11+5+2+3}{4}$$

$$= \frac{21}{4}$$

$$= 5.25$$

↓

	(12)	(34)
(12)	0	
(34)	5.25	0

min is $d_{12,34} = 5.25$

only two groups are present (12) (34) so, they will finally grouped (1234)

↓

	(1234)
(1234)	0

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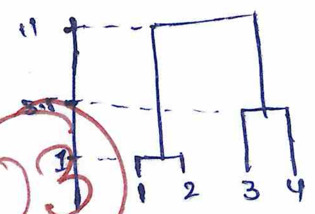
Grouping order is

$\{1,2,3,4\} \longrightarrow \{\{1,2\},3,4\} \longrightarrow \{\{1,2\},\{3,4\}\} \longrightarrow \{\{1,2,3,4\}\}$

b) Dendograms for

complete linkage



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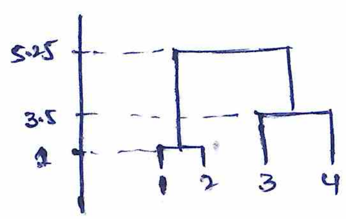
average linkage



In both methods the order of grouping is same but in average linkage final grouping is done at lower level (at 5.25) compared to higher level 11 in complete linkage

Additional Pages for Answers